



B(3rd Sm.)-Computer Science-H/DSCC-4/CCF



2024

COMPUTER SCIENCE — HONOURS

Paper : DSCC-4

(Computer Architecture and Organization)

Full Marks : 75

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

1. Answer **any five** questions :

2×5

- (a) What do you mean by pseudo code?
- (b) What is cache memory?
- (c) What is program status word?
- (d) Why is data bus bidirectional?
- (e) What is register transfer?
- (f) Explain the function of ALE and HOLD.
- (g) What is pipelining?
- (h) Define machine cycle and instruction cycle.

Answer **any three** questions.

2. Compare between DROM and SRAM.

5

3. (a) What are the role of accumulator and instruction register?

(b) What are micro operations?

3+2

4. Explain briefly the function of flag register.

5

5. Design a digital circuit that performs the four logic operations of EX-OR, EX-NOR, NOR and NAND.
Use two variables.

5

6. How does the Memory Address Register co-ordinate with the control unit to ensure correct memory addressing for data and instructions?

5

Please Turn Over

(1656)

Answer **any five** questions.

7. (a) Define bus arbitration in the context of an I/O system and explain its importance in managing shared resources.
(b) What is the function of interrupt? 6+4
8. (a) Design a digital circuit that performs the shifting operation using MUX.
(b) Write three main differences between EPROM and EEPROM.
(c) What is the function of temporary register? 5+3+2
9. (a) Write a program to evaluate the arithmetic statement : $E = (A + B)/(C - D)$ using (i) three address (ii) two address (iii) one address (iv) zero address instructions where A, B, C, D, E are memory location.
(b) What is USB? 8+2
10. (a) What is bootstrap loader?
(b) Explain with proper diagram, how does associative memory work.
(c) What is ALU? 2+6+2
11. (a) What is a memory address map?
(b) How to do the address mapping using pages in computer architecture?
(c) Mention two advantages of virtual memory? 2+6+2
12. (a) Compare and Contrast CISC and RISC architecture.
(b) What are the differences between burst mode and cycle stealing techniques of data transfer schemes? 5+5
13. Write short notes on **any two** of the following topics : 5×2
(a) Asynchronous and synchronous data transfer
(b) Different types of addressing mode
(c) Software interrupt.
14. (a) What is DRAM? How does it differ from SRAM?
(b) What are the primary memory management requirements in modern operating systems? 6+4
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